

Project Assignment: Swarm Intelligence Algorithm Implementation and Enhancement

Course: Soft Computing Optimization Techniques

Semester: Summer 2025

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Project Title: Enhancing Swarm Intelligence Algorithms with Local Search on CEC Benchmark Functions

Objective:

The objective of this project is for students to gain practical experience in the design, implementation, and evaluation of swarm intelligence algorithms. Students will:

1. Implement a standard swarm intelligence algorithm of their choice (e.g., particle swarm optimization, whale optimization algorithm, artificial bee colony, firefly algorithm, sine cosine algorithm etc.).
 2. Propose and implement a meaningful modification or enhancement to the algorithm (e.g., improved local search, hybridization, adaptive parameters).
 3. Compare the performance of the original and the modified algorithms on a standardized set of optimization problems from the CEC benchmark function suite.
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Project Tasks:

1. **Algorithm Selection and Study**
 - Select one well-known swarm intelligence algorithm.
 - Study its working principles and implementation details.
2. **Baseline Implementation**
 - Implement the original version of the selected algorithm from scratch in Python.
 - Ensure correctness by running preliminary tests on simple benchmark functions.
3. **Modification/Enhancement**
 - Design and implement a modification to the algorithm. Examples include (but are not limited to):
 - Improved local search strategy.
 - Dynamic adaptation of algorithm parameters.
 - Hybridization with another optimization method.
 - Novel update rules or initialization methods.
4. **Benchmarking**
 - Evaluate both the original and modified algorithms on a set of test functions from the **CEC benchmark suite** (e.g., CEC 2013, 2017, or 2020).

- Use a consistent number of function evaluations and independent runs (e.g., 30 runs per function).
 - Record convergence behavior and final results.
 - 5. Performance Comparison and Analysis**
 - Compare the original and modified algorithms in terms of:
 - Optimization accuracy.
 - Convergence speed.
 - Robustness (variance over multiple runs).
 - Use appropriate statistical tests (e.g., Wilcoxon signed-rank test) to assess significance.
 - 6. Report and Submission**
 - Submit a detailed report (6–10 pages) that includes:
 - Introduction and motivation.
 - Description of the original algorithm.
 - Description and rationale for the modification.
 - Experimental setup and results.
 - Performance analysis and conclusions.
 - Submit the source code with clear documentation and instructions for running experiments.
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Deliverables:

- **Report** (PDF format)
- **Source code** (well-structured and documented)
- **Result files** (e.g., CSV or plots)
- **Presentation** (defended in person in any exam term)

CEC functions to be evaluated:

F1: Bent Cigar Function

Bent Cigar Function is a commonly used optimization problem in the field of evolutionary computation. It is defined as follows:

$$f_1(x) = x_0^2 + 10^6 * \sum_{i=1}^{n-1} x_i^2$$

where x is an n -dimensional vector, and the function is subject to the constraint that $-100 \leq x_i \leq 100$ for all i .

F2: Rastrigin's Function

$$f_2(x, y) = 2A + x^2 - A * \cos(2\pi x) + y^2 - A * \cos(2\pi y)$$

where A is a constant that determines the magnitude of the oscillations in the function.

F3: High Conditioned Elliptic Function

The High Conditioned Elliptic Function is a multi-dimensional function commonly used as a benchmark function for optimization algorithms. The function is defined as:

$$f_3(\mathbf{x}) = \sum_{i=1}^n a^{\frac{i-1}{d-1}} x_i^2$$

where $\mathbf{x}=(x_1, x_2, \dots, x_n)$ is the n -dimensional input vector, a is a constant (usually set to 10^6), d is the dimensionality of the input space, and $|\cdot|$ denotes the absolute value. The function has a single global minimum at the origin, and has narrow valleys that become increasingly steep as the dimensionality of the input space increases.

F4: HGBat Function

$$f(x_1, x_2) = \left(\sum_{i=1}^2 (x_i - a_i)^2 \right)^2 + \left(\sum_{i=1}^2 (x_i - a_i) \right)^2 / 10^6$$

where $a_1 = -10$ and $a_2 = -5$.

F5: Rosenbrock's Function

$$f(x, y) = 100(y - x^2)^2 + (1 - x)^2$$

The Rosenbrock function is commonly used as a performance test for optimization algorithms, as it has many local minima and one global minimum. The global minimum is at $x=1$ and $y=1$, where $f(x,y)=0$. The function is named after Howard H. Rosenbrock, who introduced it as a test problem in his 1960 paper "An Automatic Method for Finding the Greatest or Least Value of a Function".

F6: Griewank's Function

Griewank's function is a non-convex and continuous function that is commonly used for testing optimization algorithms. It is defined as:

$$f(x) = 1 + \left(\frac{1}{4000}\right) * \sum_{i=1}^{D-1} x_i^2 - \prod_{i=1}^D \cos\left(\frac{x_i}{\sqrt{i}}\right)$$

where x_i represents the i -th element of the input vector x . The function has multiple global minima separated by large flat regions, making it a challenging optimization problem. The global minimum value of the function is $f(x)=0$, which occurs at $x=[0,0,...,0]$. The function is defined for any number of input variables.

F7: Ackley's Function

The Ackley's Function is a commonly used optimization problem that has a global minimum at $(0,0)$. It is defined as follows:

$$f(x, y) = -20e^{-0.2\sqrt{0.5(x^2+y^2)}} - e^{0.5(\cos(2\pi x) + \cos(2\pi y))} + e + 20$$

where x and y are the input variables, the function has a global minimum of 0 at $(0,0)$. The function has several local minima, which makes it a challenging problem for optimization algorithms.